COMPATIBILITY OF A TRANSFORMATION WITH ITS ADJOINT TRANSFER VIA COUNITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3774-3790, following the statement at 3758-3772.

Editorial note. The proof is complete with no omission markers.

Allowed standard prerequisites (editorial).

- Adjunction bijection and transfer of transformations. definition-adjoint and the correspondence defining beta-vee, fixed in the statement. Supplies the characterization of beta-vee as the unique transformation whose induced maps on Mor sets match those induced by beta at proof lines 3775-3778.
- Counit transformations of the two adjunctions. the unlabeled counit arrows of the square, fixed in the statement. Supplies the counit maps u_i v_i D -> D whose compatibility with beta is the claim.

Lemma 0.1. *Let $u_1, u_2 : \mathcal{C} \to \mathcal{D}$ be functors with right adjoints $v_1, v_2 : \mathcal{D} \to \mathcal{C}$. Let $\beta : u_2 \to u_1$ be a transformation of functors. Let $\beta^\vee : v_1 \to v_2$ be the corresponding transformation of adjoint functors. Then*

$$\begin{array}{ccc} u_2 \circ v_1 & \xrightarrow[\beta]{} & u_1 \circ v_1 \\ {\scriptstyle \beta^\vee}\downarrow & & \downarrow \\ u_2 \circ v_2 & \longrightarrow & id \end{array}$$

is commutative where the unlabeled arrows are the counit transformations.

Proof. This is true because $\beta^\vee_D : v_1 D \to v_2 D$ is the unique morphism such that the induced maps $\mathrm{Mor}(C, v_1 D) \to \mathrm{Mor}(C, v_2 D)$ is the map $\mathrm{Mor}(u_1 C, D) \to \mathrm{Mor}(u_2 C, D)$ induced by $\beta_C : u_2 C \to u_1 C$. Namely, this means the map

$$\mathrm{Mor}(u_1 v_1 D, D') \to \mathrm{Mor}(u_2 v_1 D, D')$$

induced by $\beta_{v_1 D}$ is the same as the map

$$\mathrm{Mor}(v_1 D, v_1 D') \to \mathrm{Mor}(v_1 D, v_2 D')$$

induced by $\beta^\vee_{D'}$. Taking $D' = D$ we find that the counit $u_1 v_1 D \to D$ precomposed by $\beta_{v_1 D}$ corresponds to $\beta^\vee_D$ under adjunction. This exactly means that the diagram commutes when evaluated on D. □

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